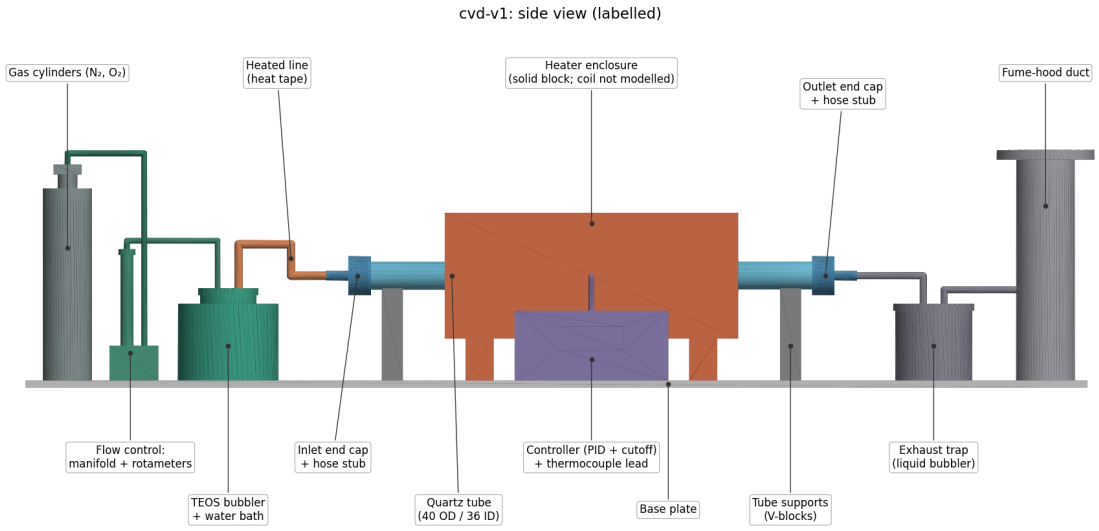
Motivation

What I would want to do in my time at HackerFab:

I would be most interested to work on the various deposition processes during my time at hackerfab. As a materials science enthusiast, I have always been intrigued by how processes like sputtering & CVD can be used for nanocoatings and precise molecular control for a wide range of functions , such as direct SiO2 deposition and Si3N4 passivation and novel innovations, such as carbon nanotubes or MEMS devices. At hackerfab, would like to build cvd-v1, a dedicated, low-cost CVD furnace, and also work on sputtering & CMOS device fabrication

Short term plan: First, I would understand the working of the existing tube furnace thoroughly, followed by designing a low-cost, atmospheric pressure CVD furnace. Once the furnace is built, I would begin with SiO2 deposition using tetraethyl orthosilicate (TEOS) onto Si substrates and compare the CVD deposited samples with the thermally grown samples by ellipsometry/profilometry. The process of building the cvd furnace would be quite similar to the tube furnace, with few important modifications. A similar kanthal based heating element would be used, but at lower temperatures (800-900 degrees celcius) and with smaller dimensions (40mm x 600mm). The quartz tube would also be need to have a future provision for being sealed, so that low pressure cvd can be implemented. For the v1 of the project, a precursor delivery line, which includes a TEOS supply and bubbler would also need to implemented. Also, for certain processes, a pure nitrogen environment would be required

Here's an LLM-generated schematic of my idea to visualize it better



Process flow:

Clean the furnace, insert a dry, pure Si substrate, seal and purge the quartz tube, check for leaks using soap water. Use the flow control to pass nitrogen gas through the quartz tube, as a cost cutting measure, if the reactions allows for it, replace pure n2 with air. Switch the carrier gas to pass through the TEOS bubbler and run it for the required time. Once the deposition is done, switch off the heating and keep passing pure n2/air through the tube as it cools. Once the tube is at room temperature, the sample can be recovered

Mid term plan: Once the TEOS oxide process is fully documented, I would shift to metal deposition, since nearly every device step depends on it. After familiarizing myself with the sputter-v2 and thermal evaporator, I would contribute to their proper characterization, i.e deposition rate against power, pressure and time, thickness uniformity across the wafer, and film cleanliness. I would then work on solving the issue of HF attacking the photoresist. A solution suggested in the docs was Chromium based hard resist layers, which is something that I would be interested to experiment with and characterize

Long term plan: My long term plan would be to help hackerfab synthesize CMOS devices. CMOS needs both n-type and p-type regions, more masking steps and many more high-temperature steps than NMOS. I would like to contribute in CVD for gate oxide deposition and source/drain masking, sputtering of the contacts and gate metal and selective doping of p and n regions

Personal Motivation: I have always been someone who loves to explore, experiment, and tinker with things. Throughout my time at IITB so far, I have explored a wide range of technical fields, including electronics and robotics, artificial intelligence and machine learning, data science, software engineering, and core materials science. I found myself to be particularly fascinated by semiconductor materials, device fabrication processes and the depths of engineering involved in them. Acting upon this interest, I have taken up a number of projects, including a summer of science on materials characterization, courses on solid state physics and carrier behaviour in materials, based on which I built a python based interactive band theory solver (link- https://homepages.iitb.ac.in/~pranav.sridhar/post.html?slug=band-theory-simulator) and currently, I work on band gap tuning and phase separation suppression in direct bandgap optical semiconductors at the OeMD lab at the MEMS department. However, through these experiences, I have come to realize that my work is inclined towards fundamental science, and is pursued largely at an individual scale. I would really like to explore beyond the fundamental properties of semiconductors and electronic materials, by understanding how they translate into devices with real use cases, and the challenges that arise in their fabrication. At hackerfab, I would have the opportunity to bridge this gap between fundamental science and engineering. I feel the constraints, such as the low costs and the fact that complex devices are to be manufactured using widely available/repurposed components would push me to develop a deeper understanding of fabrication processes, think creatively and gain

hands-on engineering experience. Something that is equally important to me, would be the community of like minded individuals, who share the same curiosity and enthusiasm as me, and hope to pursue careers in related fields. Having access to a community, would allow me to share my ideas, work upon them and work towards building something much greater than what I could do individually